Ansh Singh

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Mechanical & Simulations Engineer (M.Sc., RWTH Aachen) with hands-on experience in thermal system design, FEA/CFD simulation, and DIN/ISO-compliant prototype development. Achieved 40% energy reduction using a proprietary geothermal cooling system and 25% improvement in journal bearing reliability through novel risk-factor formulations. Additional expertise in AI/ML surrogate modelling and cryogenic systems as well as hyperscaled experimental projects.

Experience:

Global Product Manager - Kyaani GmbH — Berlin, DE

AI Product Manager — **Sept 2025 - Present**

- Directed the end-to-end development of full-stack AI engineering systems (React/TypeScript, Python/FastAPI), translating complex client requirements into structured agentic models that automate workflows.
- Architected AI assistants for computational mechanics pipelines, enabling automated task routing, root cause analysis (DMAIC), and CAD/FEM framework generation to reallocate engineering effort toward high-value design.
- Creating AI models on a closed-loop compute architecture for corporate firms that operate and run locally for maximum data privacy, data segregation and eliminating training sub-set cross-contamination of user data.

Systems/Design Engineer - Wempyre (Start-Up . Part Time) — Berlin, DE

Prototype Development - Pyrolysis Reactor and Fabrication — **April 2026 - July 2026**

- Developed Python-based data pipelines (NumPy/Pandas) to simulate thermodynamics and syngas energy yields, generating synthetic databases to optimize reactor efficiency and guide engineering decisions.
- Conducted rigorous fluidic, thermal, and structural evaluations using SolidWorks CFD/FEA to establish boundary conditions and a scalable prototype model for a full-fledged pyrolysis reactor design.
- Developed dynamic calculation models to strictly validate the reactor's geometry against European engineering standards (DIN EN 13445 for pressure vessels, DIN EN 13480 for piping, and explosion safety protocols).

Master and Mini-Thesis Work, DFG with RWTH Aachen - IMSE — Aachen, DE

Sensitivity analysis and Risk Factor Formulation for Journal Bearings — **Mar 2023 - Sept 2025**

- Conducted advanced FEM-based nodal thermal and multi-body simulations to investigate long-term journal bearing reliability, integrating physical telemetry via Texas Instruments TMP117 sensors into a temperature-based alarm system.
- Developed a predictive risk-factor framework to evaluate bearing performance under thermo-mechanical stress and mixed-lubrication conditions, achieving a 25% measurable improvement in operational reliability and efficiency.

Systems Engineer, RWTH Aachen — Aachen, DE

Neural Network Surrogate Model for a FEM-modelling in Porous Media — **Aug 2024 - Mar 2025**

- Engineered a Graph Neural Network (GNN) surrogate model using PyTorch/DGL to predict structural displacements in porous media, successfully reducing FEM simulation runtimes by 50% with near-zero accuracy loss.
- Built automated data pipelines to process massive datasets of polyurethane foam FEM simulations under dynamic loading, successfully deploying machine learning algorithms within traditional computational mechanics workflows.
- Demonstrated the potential of integrating machine learning with computational mechanics workflows to accelerate simulation-driven analysis and enable faster decision-making in engineering applications without the long computational times of normal FEA softwares.

Lead Systems Designer, Illinois Tech (Contractor for Randstad Engineering) — Chicago, IL

Geothermal Experimental Cooling Solution for Datacenters — **Jan 2019 - Oct 2020**

- Designed a complete geothermal cooling schematic for large-scale data centers, conducting full thermal analyses to optimize heat transfer, operating conditions, and long-term economic viability.
- Delivered a scalable, grid-independent green cooling solution that achieved a 40% reduction in energy consumption compared to conventional CRAC systems and a 25% reduction in prototype deployment costs.
- Deployed a cutting-edge green solution prototype using renewable geothermal cooling as a scalable alternative for energy-intensive infrastructure such as data-centres complete with cost-breakdown analysis and long-term profitability analysis.

Systems Engineer, Cryo-Chamber System, Illinois Tech (with FermiLab) — Chicago, IL

Sub-Atomic Interferometer Research Project — **Jan 2018 - Dec 2018**

- Designed a high-precision detection apparatus for sub-atomic particle measurement across a multi-component experimental assembly, integrating mechanical, thermal, and optical systems to meet strict operational tolerances. The apparatus was developed to use laser spectroscopy to test the Standard model using the weak equivalence principle.
- Engineered an industry-standard, high-precision detection apparatus for sub-atomic particle measurement, integrating mechanical, optical, and thermal systems to strict operational tolerances.
- Resolved severe thermal instability issues by developing and optimizing a superfluid liquid helium pumping system operating at near-vacuum pressures, dramatically improving temperature stability for laser spectrometers.

Zebu Group - Mobile Gaming (start-up) Mumbai, India

Co-Founder - Engineering Design **Jan 2014 - Dec 2017**

- Spearheaded the design, marketing, and cross-functional development of a puzzle-based mobile gaming application, driving the product into the top 20 rankings on iOS and Android App Leaderboards shortly after release.
- Conducted extensive market research and trend analysis to translate technical features into user-facing value propositions, culminating in the successful sale of the product IP.

Air India Engineering Services Mumbai, India

Aerospace Engineering Intern **Apr 2013 - Aug 2013**

- Worked as an engineering diagnostician on A320 and A330/350 aircrafts and Pratt-Whitney and Rolls-Royce engines between flights. Gained first hand knowledge of the working of engines and various aerodynamic and propulsion techniques in aircrafts.

Self-seeking fire extinguishing robot Houghton, MI

Team Member - Engineering, Michigan Technological University

- Engineered the mechanical chassis and fluid suppression systems using SolidWorks, balancing strict weight constraints with the structural durability and thermal resistance required for high-temperature environments.
- Collaborated with cross-functional engineering teams to package drivetrain, sensor, and fluid suppression components within strict spatial and weight constraints for maximum mobility.
- Developed autonomous navigation and thermal detection algorithms using C++ on Arduino and Python on Raspberry Pi, translating raw IR sensor data into real-time closed-loop motor control to independently locate and extinguish fire sources.

Education:

RWTH Aachen — **Aachen, DE**

Masters of Science - Mechanical Engineering, Design/Conception of Machines (Oct 2022 - Sept 2025)
Specialized in conception of machines using finite element analysis methods and modeling tools with emphasis on using neural networks and artificial intelligence as well as machine-learning algorithms for faster computation.

Illinois Institute of Technology (IIT) — **Chicago, IL, USA**

Bachelor of Science in Mechanical Engineering (2017-2019)
Research projects in cryo-engineering and experimental design with focus on Finite Element Analysis and AutoCAD modeling, predictive maintenance and failure analysis.

Skills and Tools:

Engineering Tools: CAD/CAE 2D/3D, Siemens NX 2000, Matlab Programming, Finite Element Analysis(FEA), Python, C++, Java, SolidWorks, CATIAv5, ABAQUS, JavaScript, ANSYS Fluent, Revit, AVL Excite, Simulink, COMSOL, Predictive Maintenance, Tribology, Thermal Analysis, Dynamic Systems, Fluid Dynamics, Computational Fluid Dynamics (CFD), Structural Analysis, Nonlinear Analysis, Contact Mechanics, Modal Analysis, Meshing, Mesh Independence Study, Boundary Conditions, Multibody Simulation, Verification & Validation (V&V), GD&T (Geometric Dimensioning & Tolerancing), OpenFOAM

AI and Machine-learning Tools: Neural Networks, LLM prompt engineering and API utilisation, Additive Manufacturing, PyTorch, TensorFlow, DGL, Graph Neural Networks (GNN), RAG, LangChain, NumPy, Pandas, ParaView, Surrogate Modeling, Reduced-Order Models (ROM), Scientific Machine Learning, Data Pipelines / ETL, scikit-learn, Uncertainty Quantification (UQ), Model Deployment

Project Management Techniques: IL6S, SAP, AGILE/LEAN, WBS, GANTT, Jira, Planner, Six Sigma, DMAIC, FMEA, Cross-functional Collaboration, Requirements Engineering, Stakeholder Management, Design Reviews, Technical Documentation, Root Cause Analysis (RCA)

Languages

- English (Native), Hindi (2nd language), German (Intermediate - ongoing)